NOTE

Public-key cryptography is critical to how many of the most important processes in digital identity systems work. If you are unsure about the principles of public key cryptography, I recommend Chapter 9 of my book *Learning Digital Identity* (O'Reilly Media, 2023) for a developer-level primer on the subject.

Passkeys, as specified by the FIDO Alliance, are the best, easiest-to-deploy challenge-response system available. One benefit of passkeys is the WebAuthn protocol, which integrates passkeys with browsers and JavaScript applications, making them nearly impossible to phish. If you're not familiar with passkeys, you can try them out at www.passkeys.io. The biggest drawback to passkeys is that they are unfamiliar to many users, but they are being deployed as options for authentication by many online services, including Google, Apple, Amazon, and others.

Other examples of strong, challenge-response systems in common use include SSH key authentication to servers and smart cards or USB keys for enterprise logins. These and passkeys are all based on public-private key cryptography.

Digital certificates, based on public-key infrastructure (PKI) package an identifier and public key together in a certificate that's been issued by a certificate authority. You are probably familiar with these from TLS since they're used to link the public key used to encrypt HTTPS traffic with an identifier in the form of a domain name. But they can also include attributes about the identifier, and the issuer can (for a fee) conduct more thorough identity proofing than is commonly done for certificates used in TLS. This makes them very flexible because they don't necessarily require pre-registration for the requester to prove they are in control of an identifier. So long as the authentication service trusts the certificate issuer, certificates can provide a rich set of attributes about the identifier without registration.